PROJECT PRESENTATION | 7TH SEMESTER

ManologAI

A Unified Personal Life Analytics Platform & Behavioral Insight Engine

PROJECT SUPERVISOR

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SUBMITTED BY (STUDENTS)

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BACKGROUND & MOTIVATION

The Problem: The Fragmented Quantified-Self

B.Tech IT (Sem VII)

⚠️ Current Industry Challenges

- **Siloed Life Data:** Users record habits in one app, journals in Notion, mood in Daylio, and sleep in health trackers without interoperability.
- **Zero Cross-Domain Correlation:** Isolated apps cannot answer critical lifestyle questions: *"How does sleeping under 6.5 hours impact my daytime stress and work output?"*
- **Absence of Data Integrity:** Most self-tracking apps permit retroactive editing, leading to cognitive bias and inflated streaks.
- **High Logging Friction:** Dispersed workflows cause user burnout within 2–3 weeks.

🎯 Project Objectives

- **Unified Behavioral Intelligence:** Merge qualitative journaling with quantitative time, expense, and habit tracking in a single ecosystem.
- **Personal Data Scientist:** Implement statistical algorithms to expose hidden correlations between sleep, screen time, habits, and emotional state.
- **Integrity Enforcement:** Implement an immutable 24-hour write lock to prevent retroactive revisionism.
- **Frictionless UX:** Native Progressive Web App (PWA) with VAPID web push reminders under a sub-2-minute daily routine.

CORE CAPABILITIES

Functional Modules & System Features

📖 Daily Journal & Mood

- Single entry per day with auto-save and calendar navigation.
- Tagging system & day rating (0–10).
- Emotion logger mapping daily sentiments for longitudinal tracking.
- Client-side encrypted PDF export.

⏱️ Time & Habits Engine

- Rapid logging: Sleep, Screen Time, Work/Study hours, and Expenses.
- Sum integrity validation (Daily hours ≤ 24).
- Flexible habit schedules (Daily, Weekly, Custom Day intervals).
- GitHub-style contribution heatmaps.

⚖️ Wheel of Life & Planner

- Evaluates 8 life dimensions (Health, Career, Finance, Relationships, etc.).
- Structured Pads for Goals, Books, Ideas, and To-Dos.
- Weekly timetable creator with real-time overlap & conflict detection.

SYSTEM ENGINEERING

Full-Stack System Architecture

High-Level Design

PRESENTATION TIER

React 19 PWA

Vite 8, Tailwind v4, Recharts, Framer Motion, Service Worker

SECURITY & PROXY

Gateway / Middleware

Vite Proxy, Dual JWT verification, HTTP-only Cookie rotation

APPLICATION TIER

Express 5 REST API

Domain routers, Statistical analytics engine, Push scheduler

DATA TIER

MongoDB & Cloudinary

Compound index schemas, Multi-device sessions, Media assets

Architectural Principles & Pipeline

- **Decoupled Micro-layering:** Client-side rendering backed by stateless REST endpoints ensuring fast Time-To-Interactive (TTI).
- **Compound Constraints:** Database models enforce `{ user: 1, date: 1 }` unique compound indices to eliminate race conditions and duplicates.
- **Proactive Push Scheduler:** Node.js background worker executes 60s cron evaluation with automatic dead-subscription purging (HTTP 410).
- **Dual-Layer Fallback:** Seamless handoff between OS Web Push and in-browser Notification API.

SYSTEM ARCHITECTURE & DATAFLOW

End-to-End Architecture Diagram & Dataflow Pipeline

1. Client Layer — PWA

- **React 19 & Vite 8:** Reactive UI & state stores
- **PWA Service Worker:** Cache runtime & background sync
- **VAPID Push Listener:** Subscribes via PushManager

→ `HTTPS / COOKIE`

2. Ingestion Gateway — SECURITY

- **Reverse Proxy:** Route dispatching & CORS
- **Dual JWT Auth:** HttpOnly cookie token rotation
- **24-Hour Guard:** Temporal write-lock validation

→ `SANITIZED REQ`

3. Analytics & Services — EXPRESS 5

- **Domain Routers:** Diary, Habits, Time, Emotions
- **Correlation Engine:** Pearson *(r)* & Habit Deltas (Δ)
- **Web Push Daemon:** 60s cron evaluation & dispatch

→ `MONGOOSE ODM`

4. Persistence Tier — STORAGE

- **MongoDB:** Compound key `{ user: 1, date: 1 }`
- **State Collections:** Entries, Emotions, Ratings, Habits
- **Cloudinary:** CDN profile asset hosting

📝 Log Ingestion & Integrity Flow

1. **User Submission:** User submits daily diary, emotion, habit, or time allocation entry.
2. **Integrity Check:** Middleware validates entry date ≤ current day and within 24h editing window.
3. **Atomic Upsert:** MongoDB updates record matching `{ user, date }` with zero duplicate risk.

📊 Behavioral Analytics Flow

1. **Query Window:** Client requests 7/30/90-day time-series and habit history over REST.
2. **Quantization & Math:** Engine computes Pearson correlation coefficient *(r)* and habit delta Δ.
3. **Visualization:** Recharts renders reactive scatter plots, radar charts, and trend lines.

🔔 VAPID Push Notification Flow

1. **Subscription Handshake:** Browser generates endpoint & auth keys via VAPID Public Key.
2. **Scheduler Match:** Backend 60s cron detects due reminders (Daily Log, Habits, Check-in).
3. **Payload Delivery:** Web-Push RFC 8291 payload awakens Service Worker on device lockscreen.

TECHNICAL INNOVATIONS

Key Technical Innovations & Engineering Highlights

Core Innovations

🔒 24-Hour Integrity Lock Engine

- **Strict Temporal Lock:** Diary, emotion, and time logs enforce an immutable 24-hour write lock to prevent revisionist bias.
- **Middleware Guard:** Future-dated entries and retroactive updates are blocked at the Express API layer.
- **Dataset Fidelity:** Guarantees unpolluted, realistic behavioral logs essential for reliable statistical inference.

🧠 Cross-Domain Correlation Engine

- **Pearson Correlation *(r)*:** Calculates statistical correlation between lifestyle metrics (sleep/screen time) and mood scores.
- **Day-of-Week Clustering:** Surfaces weekly emotional dips, stress spikes, and habit drop-off patterns.
- **Counterfactual Habit Deltas:** Compares 8-domain "Wheel of Life" scores on days habits are completed vs. skipped.

🛡️ Robust Session & Token Rotation

- **Dual-Token Auth:** Short-lived access token (15 min) paired with 7-day refresh token rotation.
- **Cookie Hardening:** Stored in `httpOnly`, `SameSite: Lax` cookies preventing client-side script theft (XSS).
- **Multi-Device Pruning:** Active device tracking with automated token reuse detection and instant session revocation.

🔔 VAPID Web Push & Mobile PWA

- **Standards-Based VAPID:** Direct RFC 8291/8292 web-push protocol implementation with zero proprietary SDKs.
- **Installable Standalone PWA:** Native full-screen mobile app shell with custom splash screen and offline service worker.
- **Automated Dead-Endpoint Purging:** Background scheduler prunes expired subscriptions upon receiving HTTP 410.

Technology Stack & Tooling

💻 Frontend Stack

- **React 19:** Latest React runtime leveraging modern hooks and component patterns.
- **Tailwind CSS v4:** Next-gen CSS engine with optimized build pipelines.
- **Vite 8:** High-performance ES-module build & HMR tool.
- **Recharts v3:** Dynamic SVG visualizations (Radar, Area, Bar, Scatter).
- **Framer Motion:** Smooth physics-based UI transitions.

⚙️ Backend Stack

- **Node.js & Express 5:** Scalable, async RESTful API framework.
- **Mongoose 9.3:** Object Data Modeling with schema validation and pre-save hooks.
- **JWT & Bcrypt:** Industry-standard cryptographic auth and salt hashing.
- **Web-Push:** Direct implementation of RFC 8291/8292 VAPID push protocol.
- **Multer & Cloudinary:** Multipart media pipeline for profile imagery.

🗄️ Database & Delivery

- **MongoDB:** Document datastore supporting flexible, schema-driven analytics.
- **Vite PWA / Workbox:** Service worker caching strategies & manifest delivery.
- **VAPID Protocol:** Native browser push notification gateway.
- **jsPDF:** In-browser document synthesis for instant PDF reporting.

SYSTEM VERIFICATION

Demonstration & Test Results

✅ End-to-End Automated Test Suite

16 / 16 Integration Test Scenarios Passed (100% Success Rate)

- **Authentication & Profile:** Verified user registration, token cookie issuance, and profile retrieval.
- **Logging Engine:** Successfully logged daily diary, emotions, time tracking, life ratings, and habits.
- **Conflict Resolution:** Verified timetable schedule block conflict detection.
- **Push Notifications:** Tested VAPID subscription endpoints and background scheduler.

📊 Functional Highlights

- **Gamified Habit Badging:** Tiered milestones (Bronze I/II, Silver I/II, Gold, Diamond) incentivize user persistence.
- **Responsive Ergonomics:** Unified layout rendering desktop sidebar alongside an adaptive mobile tab bar.
- **Zero Console Defects:** Production build validated cleanly with 0 ESLint errors and 0 runtime browser warnings.

CONCLUSION

Project Summary & Future Roadmap

📌 Conclusion

- Successfully built and deployed **ManologAI**, bridging the gap between subjective reflections and quantitative behavioral metrics.
- Validated that a single, unified data model eliminates self-tracking friction and unlocks deep correlation insights.
- Proved the efficacy of the 24-hour integrity lock for maintaining clean personal data science datasets.

🚀 Future Scope

- **Local LLM Integration:** Integrate privacy-first on-device AI for automated weekly reflection summaries and proactive lifestyle nudges.
- **Wearables API Sync:** Integrate Google Health Connect and Apple HealthKit for automated passive sleep and workout logging.
- **CRDT Offline Sync:** Implement Conflict-free Replicated Data Types for peer-to-peer multi-device synchronization.

Thank You

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